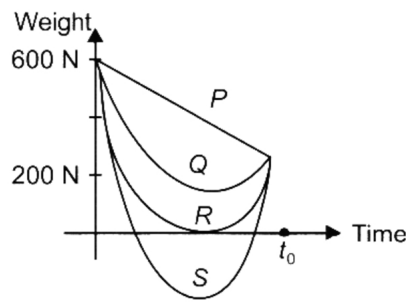


other objects in the sky. Which part of figure best represents the weight (net gravitational force) of the passenger as a function of time?



- a) S
b) R
c) P
d) Q

9. A stone dropped from a certain height takes 4 second to reach the ground. The height is [1]

- a) 19.6 m
b) 39.2 m
c) 78.4 m
d) 156.8 m

10. Match the column I with column II and select the correct option from the given codes. [1]

Column I	Column II
(a) g at height h	(i) $g \left(1 - \frac{2h}{R} \right)$
(b) g at depth h	(ii) $\frac{GM}{R^2}$
(c) g_{poles}	(iii) $g \left(1 - \frac{h}{R} \right)$
(d) g_{surface}	(iv) Maximum

- a) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
b) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
c) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)
d) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

11. A tank 2 m high is half filled with water and then filled to the top with oil of density 0.60 g/cc. What is the pressure at the bottom of the tank due to these liquids? (Take $g = 10 \text{ m s}^{-2}$) [1]

- a) $1.6 \times 10^2 \text{ N m}^{-2}$
b) $1.6 \times 10^4 \text{ N m}^{-2}$
c) $0.8 \times 10^4 \text{ N m}^{-2}$
d) $0.8 \times 10^2 \text{ N m}^{-2}$

12. Earth goes round the Sun with [1]

- a) same velocity
b) varying K.E.
c) same momentum
d) varying velocity

13. The weight of an object is [1]

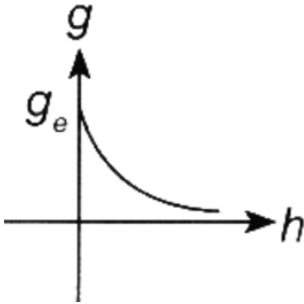
- a) Equal on both earth and Moon
b) Lesser on earth and Greater on earth
c) None of these
d) Greater on earth and lesser on Moon

14. A given solid is weighted in air using a spring balance. It is then weighed separately by immersing it fully first in vessel containing tap water and then in a vessel containing salt solution. The reading of the spring balance would be: [1]

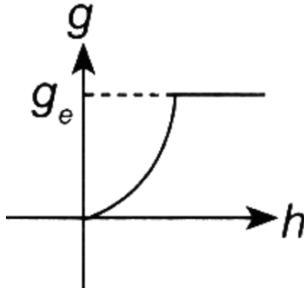
- a) least in water
b) equal in all the three cases

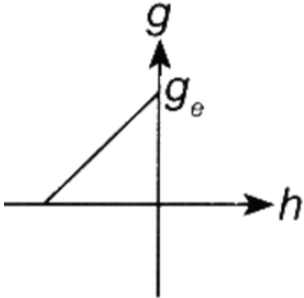
- c) least in air d) least in salt solution
15. The amount of heat energy lost by hot body in a given time depend on: [1]
 a) all of these b) the area of its surface
 c) the temperature of its surface d) the nature of the surface
16. Priya buys a bag of sugar of weight W at a place on the equator. She takes it to Antarctica. The weight of a sugar bag at Antarctica [1]
 a) Remains same as the weight at a place on equator b) Cannot be said
 c) Decreases d) Increases
17. The atmosphere is held to the earth by [1]
 a) wind b) clouds
 c) earth's magnetic field d) gravity
18. Acceleration due to gravity (due to the shell) at any point inside the shell is [1]
 a) zero as long as no other mass is outside the shell. b) cannot be found
 c) zero always d) zero as long as no other mass is inside.
19. The height at which the acceleration due to gravity becomes $\frac{g}{9}$ in terms of R , where g is the acceleration due to gravity on the surface of the Earth is (R is the radius of the Earth) [1]
 a) $\frac{R}{2}$ b) $\frac{R}{\sqrt{2}}$
 c) $2R$ d) $\sqrt{\frac{2}{R}}$
20. Acceleration due to gravity varies with [1]
 a) shape of the planet b) All of these
 c) height d) depth
21. The gravitational force between two masses kept at a certain distance is p N. The same two masses are now kept in water and the distance between them is kept same. The gravitational force between these two masses in water is q N. Then [1]
 a) $p > q$ b) $p < q$
 c) $p = q$ d) None of these
22. Acceleration due to gravity is [1]
 a) All of these b) Inversely proportional to square of radius for all points on the surface of the planet.
 c) Dependent directly on the radius of the planets under comparison are made of same material. d) Dependent on mass of the planet for points outside the planet.
23. While performing an experiment a solid was immersed fully in water and it was observed that the solid has suffered an apparent loss in its weight. This apparent loss in weight of the solid must be equal to the: [1]

- a) weight of water displaced by the solid b) weight of the solid in water
c) weight of the solid in air d) weight of water filled in the vessel
24. An object weights 10 N in air. When immersed fully in water, it weighs only 8 N. The weight of the liquid displaced by the object will be [1]
a) 12 N b) 8 N
c) 2 N d) 10 N
25. A body is suspended from a spring balance kept in a satellite. The reading of the spring balance is W_1 when the satellite goes in an orbit of radius R and W_2 when it goes in an orbit of radius $2R$. Which of the following is correct? [1]
a) $W_1 \neq W_2$ b) $W_1 > W_2$
c) $W_1 < W_2$ d) $W_1 = W_2$
26. Acceleration due to gravity on the surface of the earth is the greatest [1]
a) uniform at all places b) at equator
c) at poles d) at 23.6° latitude
27. Which of the following graphs shows an approximate variation of g with height h above the Earth's surface? [1]
- a)

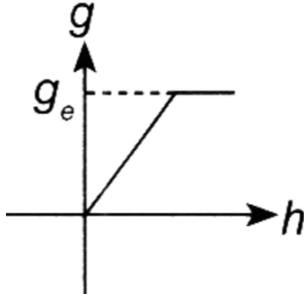


b)


- c)



d)


28. Two particles are placed at some distance. If the mass of each of the two particles is doubled, keeping the distance between them unchanged, the value of gravitational force between them will be [1]
a) Unchanged b) $\frac{1}{4}$ times
c) $\frac{1}{2}$ times d) 4 times
29. The weight of an object at the center of the earth of radius R is [1]
a) $\frac{1}{R^2}$ times the weight at surface of the earth b) zero
c) infinite d) R times the weight at the surface of the earth
30. The value of gravitational constant depends upon [1]

- a) temperature of the atmosphere b) masses
c) distance between the masses d) none of these
31. The weight of an object has S.I. unit of: [1]
a) Newton b) N/kg
c) kg/N d) kg
32. A boy is whirling a stone tied with a string in a horizontal circular path the string breaks, the stone [1]
a) will continue to move in the circular path b) will move along a straight line towards the centre of the circular path
c) will move along a straight line perpendicular to the circular path away from the boy. d) will move along a straight line tangential to the circular path
33. A student lowers a body in a liquid filled in a container. He finds that there is a maximum apparent loss in weight of the body when: [1]
a) it is partially immersed and also touches the sides of the container. b) it is partially immersed in the liquid.
c) it just touches the surface of the liquid. d) it is completely immersed in the liquid.
34. The mass of body is 10 kg at a place where $g = 10 \text{ m/s}^2$, its weight is [1]
a) 100 N b) 1000 N
c) 10 N d) 1 N
35. A heavy cylinder of length 1 is slowly taken out of a dense liquid. The weight felt as it is taken out of the liquid [1]
a) increases till it attains the weight in air b) decreases as it comes out
c) increases as it comes out d) will remain the same
36. A girl stands on a box having 60 cm length, 40 cm breadth and 20 cm width in three ways. In which of the following cases, pressure exerted by the brick will be [1]
a) maximum when width and length form the base b) maximum when length and breadth form the base
c) maximum when breadth and width form the base d) the same in all the above three cases
37. When a spaceship is two earth radii distant from the centre of the earth, its gravitational acceleration is [1]
a) 4.9 ms^{-2} b) 19.6 ms^{-2}
c) 2.45 ms^{-2} d) 9.8 ms^{-2}
38. The physical quantity in which apparent loss is observed when a solid object is immersed in water. [1]
a) Volume b) Density
c) Mass d) Weight
39. Variation of acceleration due to gravity g with distance r from the centre of earth with ($r > R_e$) is best given as [1]
a) b) $g \propto r$

$$g \propto r^2$$

c) $g \propto \frac{1}{r}$

d) $g \propto \frac{1}{r^2}$

40. There is no atmosphere on moon as [1]
- a) it gets light from sun. b) it is closer to the earth.
- c) it revolves round the earth. d) the gases have less requirement of velocity or energy to escape from its surface.
41. A 60 kg man weight W kg on the moon. W is [1]
- a) 10 kg b) 60 kg
- c) 5 kg d) 70 kg
42. If the earth stops rotating the value of 'g' at the equator will [1]
- a) none of these b) remain same
- c) decrease d) increase
43. A ball is released from the top of a tower of height h metres. It takes T seconds to reach the ground. What is the position of ball above the ground in $\frac{T}{5}$ seconds? [1]
- a) $\frac{24}{25}h$ m b) $\frac{h}{25}$ m
- c) 25 hm d) 24 hm
44. Which of the following is not correct? [1]
- i. If we drop a 10 lb iron ball and 1 lb aluminium ball from the top of a high platform inside a mine, then both will reach the floor at the same time.
- ii. The gravitational force of attraction between the two bodies increases when their masses are increased and distance is reduced.
- iii. We do not observe small objects moving towards each other because frictional force balances the gravitational pull.
- iv. When a body is thrown up, the acceleration due to gravity at the topmost point is zero because velocity of the body is zero.
- a) Statement (iv) is correct. b) Statement (iii) is correct.
- c) Statement (ii) is correct. d) Statement (i) is correct.
45. The density of tap water is less than that of sea water. If an object is immersed completely in both one by one, its loss in weight will be : [1]
- a) Zero in sea water but not so in tap water. b) More in sea water.
- c) Equal in the two cases. d) Less in sea water.
46. If the volume of a body is V and half of it is immersed in the liquid, the upthrust will be proportional to: [1]
- a) Difference in density of the body and half the density of the liquid. b) Half the volume
- c) Volume d) Double the volume
47. Law of gravitation gives the gravitational force between [1]

- a) any two bodies having some mass b) two charged bodies only
c) the earth and Sun only d) the earth and a point mass only
48. An apple and a stone dropped from certain height accelerates [1]
a) depending on density b) depends on the position of the SUN
c) equally d) differently
49. If we want to determine the volume of a solid by immersing it in water, the solid should be [1]
a) heavier than water and insoluble in it b) lighter than water
c) heavier than water d) insoluble in water
50. The ratio of weight to mass of the body is _____. [1]
a) zero b) 9.8
c) 1.0 d) 10
51. The upward force experienced by a body when submerged in a fluid is called [1]
a) Buoyant force b) Muscular force
c) Liquid force d) Frictional force
52. The weight of a body felt in tap water and salty water are W_A and W_B respectively, then : [1]
a) $W_A = W_B$ b) $W_A > W_B$
c) $W_A < W_B$ d) $W_B = 2W_A$
53. In a tank having 'h' height of liquid, two points at a depth $\frac{3h}{4}$ from free surface have pressure P_A and P_B . When [1]
point A is closer to the wall than B. Then
a) $P_A > P_B$ b) $P_A < P_B$
c) $P_A = P_B$ d) Depends on density
54. If a planet existed whose mass was twice that of Earth and whose radius 3 times greater, how much will a 1 kg [1]
mass weigh on the planet?
a) 1.1 N b) 25 N
c) 2.17 N d) 5 N
55. If cross - sectional area of an object is more than the pressure applied by the external force is :- [1]
a) Less b) More
c) None of the above. d) Remains same
56. A heavier and a lighter body have equal momentum, then [1]
a) heavier will have more K.E. b) lighter will have more K.E.
c) they will have equal K.E. d) K.E. will be independent of momentum
57. The value of quantity G in the law of gravitation [1]
a) is independent of mass and radius of the earth b) depends on both mass and radius of the earth

- c) depends on mass of the earth only d) depends on radius of the earth only
58. The least value of apparent weight of a body in a fluid is [1]
 a) = 0 b) depends on the density of solid and fluid
 c) < 0 d) > 0
59. Two objects of different masses falling freely near the surface of the moon would [1]
 a) have different acceleration b) have same velocities at any instant
 c) undergo a change in their inertia d) experience forces of same magnitude
60. Increase in momentum by 20% will increase the K.E. by [1]
 a) 10% b) 20%
 c) 40% d) 44%
61. A mass weighing W can float if the upthrust U is related as [1]
 a) $U > W$ b) $U \geq W$
 c) $U = W$ d) $U < W$
62. A ball is thrown vertically upward, acceleration due to gravity is [1]
 a) In the direction same as the direction of motion.
 b) In the direction opposite to the direction of motion.
 c) Increases as it comes down.
 d) Becomes zero at the highest point.
63. In the relation $F = \frac{Gmm}{d^2}$, the quantity G [1]
 a) is greater at the surface of the earth b) depends on the value of g at the place of observation
 c) is universal constant of nature d) is used only when the earth is one of the two masses
64. Acceleration due to gravity on the surface of the moon is $\frac{1}{6}$ th of its value on the surface of the earth as they have different [1]
 a) radius b) All of these
 c) density d) mass
65. Two bodies, one held 1 m vertically above the other, are released simultaneously and fall freely under gravity. After 2 second, the relative separation of the bodies will be [1]
 a) 4.9 m b) 19.6 m
 c) 9.8 m d) 1 m
66. If the weight of an object in air and a dense liquid are W_a and W_l respectively, then upthrust [1]
 a) $U = W_l + W_a$ b) $U = 2W_l$
 c) $U = W_a - W_l$ d) $U = W_l - W_a$
67. A coil of wire with cross-sectional area 0.50 mm^2 weighs 75N in air and 65N in water. The length of the coil in cm is (Take $g = 10 \text{ m/s}^2$) [1]

a) $\frac{10^2}{0.50}$

b) $\frac{10^6}{5}$

c) $\frac{10^5}{0.0050}$

d) $\frac{10^2}{50}$

68. Any object having sufficient positive energy can escape from the gravitational pull of the earth as [1]
 a) K.E. will be more to escape the gravitational potential energy b) K.E. is negative otherwise
 c) Negative energy means it is bound d) P.E. is positive for total energy being positive
69. Even though stone also attracts earth towards itself, the earth does not move: [1]
 a) Force exerted by stone is less b) Because of lesser mass of stone
 c) Because of greater mass of earth d) Force exerted earth is large
70. A wooden block is floating in water contained in a beaker. During the free fall of the beaker, the upthrust on the block will be [1]
 a) Slightly less than the weight of the block. b) Equal to the weight of the block
 c) Zero d) Slightly more than the weight of the block.
71. In a spring balance the space between 0 and 25 g marks is divided into 10 equal parts. The least count of spring balance is: [1]
 a) 15 g wt b) 2.5 g wt
 c) 25 g wt d) 0.25 g wt
72. The value of acceleration due to gravity [1]
 a) increases from pole to equator b) is least on equator
 c) is same on equator and poles d) is least on poles
73. A body at rest having uniform acceleration is [1]
 a) possible only near the sun. b) any body thrown vertically up against gravity.
 c) can be released only on the moon. d) non-existing
74. **Statement 1:** The weight of the body becomes zero at the centre of the earth. [1]
Statement 2: As the distance from centre of earth decreases, acceleration due to gravity also decreases.
 a) Both statement 1 and statement 2 are true but statement 2 is not the correct explanation of statement 1. b) Both statement 1 and statement 2 are true and statement 2 is the correct explanation of statement 1.
 c) Both statement 1 and statement 2 are false. d) Statement 1 is true but statement 2 is false
75. **Statement 1:** A pendulum can vibrate in an artificial satellite. [1]
Statement 2: Inside the satellite, acceleration due to gravity is zero.
 a) Statement 1 is true but statement 2 is false. b) Statement 1 is false but statement 2 is true.
 c) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1. d) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

